

RLVR-Sci Experiment Report

1. Research Overview

This project implements **Reinforcement Learning from Verifiable Reward (RLVR)** for **scientific hypothesis generation** — specifically, teaching LLMs to generate novel chemical molecules (as SMILES strings) with targeted physical properties (HOMO-LUMO gap).

The key innovation is a **fast surrogate verifier** (an MLP trained on QM9 data) that approximates expensive DFT quantum-chemistry calculations, making it feasible to run inside an RL training loop. The RL algorithm is **Group Relative Policy Optimization (GRPO)** via HuggingFace `trl`.

flowchart LR

```
A["QM9 Dataset<br/>133,885 molecules"] --> B["Surrogate MLP<br/>Morgan FP → Gap"]
B --> C["GRPO Reward Signals"]
C --> D["LLM Policy<br/>(LoRA fine-tuning)"]
D --> E["generates SMILES"]
E --> F["5 reward functions"]
F --> D
```

Reward Function Architecture

#	Reward	Weight	Description
1	Validity	1.0	RDKit parse check — valid SMILES = +1.0, invalid = -1.0
2	Surrogate Gap	1.0	Normalized predicted HOMO-LUMO gap from the MLP verifier
3	Novelty	0.5	Penalizes Tanimoto similarity ≥ 0.85 to QM9 training molecules
4	Diversity	0.3	Rewards structurally diverse completions within each prompt group
5	Conditional Target	0.5	For property-conditioned prompts, rewards meeting the requested gap threshold

2. Dataset & Surrogate Verifier

QM9 Dataset

Metric	Value
Source	DeepChem QM9 CSV
Total molecules	133,885
Invalid fingerprints	0
Fingerprint dims	133,885 × 2048 (Morgan FPs, radius=2)
Gap range	0.0246 – 0.6221 Hartree

Gap mean $\pm$ std	0.2511 $\pm$ 0.0475 Hartree
Volume file	qm9_processed.csv (2.9 MiB) + qm9_fingerprints.npy (261.5 MiB)

Surrogate Verifier Performance

Metric	Value
Test R ²	0.9488
Test MSE	1.168 $\times 10^{-4}$
R ² gate threshold	0.70
Architecture	MLP (256, 128) with early stopping
Model file	surrogate_gap_model.joblib (12.8 MiB)

[!TIP] R² = 0.949 is excellent — the surrogate explains ~95% of the variance in HOMO-LUMO gap from fingerprints alone, far exceeding the 0.70 gate.

3. LoRA Configuration (All Runs)

All three GRPO runs used identical LoRA adapters:

Parameter	Value
PEFT type	LoRA
Rank (r)	16
Alpha	32
Dropout	0.05
Bias	none
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Task type	CAUSAL_LM
DoRA	false
RSLoRA	false

4. Training Results

4.1 Smoke Test — Qwen2.5-0.5B-Instruct (10 steps)

Config: 50 prompts · 10 steps · batch=4 · 4-bit NF4 · L4 GPU

Metric	Step 10 (train)	Step 10 (eval)
Total Reward	—	-0.777

Reward std	—	0.447
Loss	−0.066	−0.011
Validity	—	−0.900
Surrogate gap	—	+0.108
Novelty	—	−0.050
Diversity	—	+0.050
Conditional gap	—	NaN (skipped)
Completion length	—	39.3 tokens
Clip ratio	—	0.0

[!NOTE] The smoke test only ran 10 steps — too few for meaningful learning. Validity at −0.9 means 95% of outputs were invalid SMILES. This is expected for a 0.5B model with negligible training. The test verified pipeline correctness, not model quality.

4.2 Integration Test — Qwen2.5-0.5B-Instruct (60 steps)

Config: 180 prompts · 60 steps · batch=4 · 4-bit NF4 · L4 GPU

Training Reward Progression

Step	Total Reward	Validity	Surrogate	Novelty	Diversity	Grad Norm	Compl. Len
10	−0.170	−0.613	+0.346	+0.103	+0.118	2.36	53.0
20	+0.008	−0.525	+0.396	+0.109	+0.173	3.18	62.1
30	+0.466	−0.375	+0.665	+0.200	+0.119	5.95	58.1
40	+0.955	−0.163	+0.858	+0.274	+0.202	9.77	63.0
60	+1.714	+0.113	+1.213	+0.395	+0.195	6.03	71.1

Eval at Step 50

Metric	Value
Eval Reward	+1.001
Eval Validity	−0.167
Eval Surrogate	+0.847
Eval Novelty	+0.266
Eval Diversity	+0.194
Eval Completion Length	70.6 tokens

Trend Analysis

```
xychart-beta
  title "Qwen 0.5B – Reward Progression (60 steps)"
  x-axis "Step" [10, 20, 30, 40, 60]
  y-axis "Score" -1 --> 2
  line "Total Reward" [-0.17, 0.008, 0.466, 0.955, 1.714]
  line "Validity" [-0.613, -0.525, -0.375, -0.163, 0.113]
  line "Surrogate" [0.346, 0.396, 0.665, 0.858, 1.213]
```

[!IMPORTANT] **Clear learning signal detected.** Over 60 steps:

- **Validity** flipped from $-0.61 \rightarrow +0.11$ (model learned to produce valid SMILES)
- **Surrogate gap** nearly quadrupled: $0.35 \rightarrow 1.21$ (model learned to optimize for higher HOMO-LUMO gap)
- **Novelty** improved steadily: $0.10 \rightarrow 0.40$ (model generates increasingly novel molecules)
- **Total reward** grew monotonically from $-0.17 \rightarrow +1.71$
- Gradient norm peaked at step 40 (9.77) then stabilized at 6.03 — healthy training dynamics

4.3 Integration Test — Mistral-7B-Instruct-v0.2 (25 steps)

Config: 100 prompts · 25 steps · batch=4 · 4-bit NF4 · L4 GPU · 24 GiB memory

Training Reward Progression

Step	Total Reward	Validity	Surrogate	Novelty	Diversity	Grad Norm	Compl. Len
10	+0.051	-0.438	+0.307	+0.220	+0.138	1.81	91.2
20	+0.536	+0.088	+0.016	+0.429	+0.332	2.64	111.7

Eval at Step 25

Metric	Value
Eval Reward	+1.059
Eval Validity	+0.450
Eval Surrogate	+0.061
Eval Novelty	+0.606
Eval Diversity	+0.384
Eval Completion Length	117.6 tokens
Eval Runtime	505.5s

[!IMPORTANT] **Mistral-7B learns faster than Qwen-0.5B.** Key observations:

- **Validity flipped positive by step 20** (+0.088 train, +0.45 eval) — the larger model already produces valid SMILES majority of the time after just 20 steps, vs ~50 steps for Qwen

- **Novelty is the strongest reward** at +0.606 eval — Mistral generates structurally distinct molecules far from QM9
- **Diversity is high** at +0.384 eval — within-group molecular variety
- **Surrogate gap is low** (+0.061 eval) — the model optimizes for novelty/validity rather than high HOMO-LUMO gap at this early stage. This is expected: the model first learns to output valid, novel SMILES before learning to maximize the specific property
- **Longer completions** (117.6 tokens vs 70.6 for Qwen) — Mistral generates more verbose outputs with richer molecular descriptions
- **Gradient norms are lower** (1.8–2.6 vs 2.4–9.8 for Qwen) — more stable optimization in the larger model

5. Cross-Model Comparison

Metric	Qwen 0.5B (60 steps)	Mistral 7B (25 steps)
Final eval reward	+1.001	+1.059
Eval validity	−0.167	+0.450
Eval surrogate	+0.847	+0.061
Eval novelty	+0.266	+0.606
Eval diversity	+0.194	+0.384
Steps to valid SMILES	~50	~20
Adapter size	33.6 MiB	160.1 MiB
Completion length	70.6 tokens	117.6 tokens

*[!TIP] **Mistral-7B reaches higher overall reward in fewer steps** and excels at validity, novelty, and diversity. Qwen-0.5B excels at surrogate gap optimization (producing high-gap molecules). With more training steps, Mistral should learn to also optimize the surrogate gap — making it the clear candidate for the full 500-step production run.*